Choosing internal storage

Drive Types

M.2

2280 (22x80mm)

2230 (22x30mm)

- Newest tech (Solid state) NVMe
- Different physical sizes
- Slot into mobo
- Mobo needs M.2 slots accepting correct size
- Performance increases with price and PCIe Gen
- Theoretically upto 80 Gbps
- PCIe Gen 5 drive will run slower in Gen 4 mobo
- Can get hot - Heatsinks advisable
- Expensive

SSD

2.5"

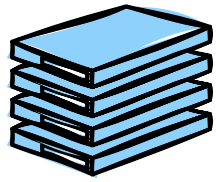
- Solid State Drive
- Connects over SATA
- Mobos have 6 or 8 slots usually
- Midrange price per TB
- Performance limited by SATA III spec
- 6 Gbps Max

HDD

5.25"

- Hard Disk Drive
- Moving parts create noise
- Connects over SATA
- Mobos have 6 or 8 slots usually
- Best price / TB
- Performance limited by SATA III spec
- Any by physics/RPM
- Fastest drive is approx 4 Gbps
- Data recovery is possible

RAID Arrays

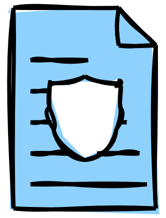


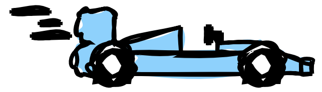
HDD's fail over time

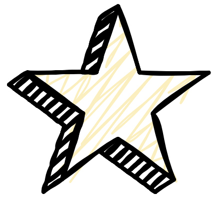
SSD's have a lifespan....

RAID = Redundant Array of Inexpensive Disks

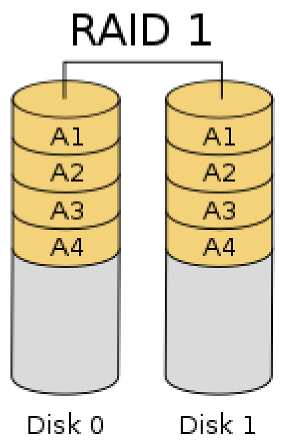
2 or more drives working together to either give data security or speed







RAID 1 Mirror



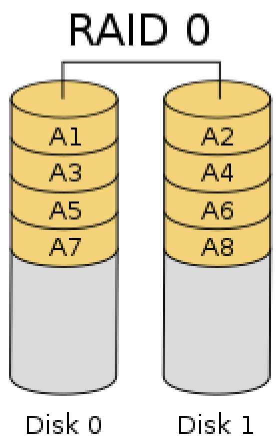
Min 2 drives

Data mirrored (copied) across all

Drives can fail and be restored

No performance benefit

RAID 0 Striped



Min 2 drives

Data striped (split) between drives

One drive fails, array fails

Performance is increased

With the advent of M.2 drives, speed is less necessary for most users except for extreme circumstances (8k video editing)

RAID 5 = Min 3 Drives, one can fail, performance is increased

RAID 10 (or 1+0) = Min 4 Drives, better performance and fault tolerance than RAID 5

Storage Recommendations

On a Budget

Windows 11 Minimum is 64GB

HDD's are the cheapest option

Gamer

Pure gaming performance.... M.2

A PS5 comes with 1TB

AAA games demand a certain drive speed

WFH Pro

64GB to 1TB+

Multiple Drives RAID1 Mirror

SSD or M.2

Creative Pro / Engineer

Check your use case, video?